

ATO → S4 PO Creation, Change Document & Audit Log

Step-by-Step Test Cases

SAP S/4HANA Public Cloud | BTP ATO SaaS | GxP Compliance

System: QJ6-001 | Package: CGTO_S4HC_POC | Date: 2026-06-26

End-to-End Test Flow Overview

#	Test Case	System	Status
TC-01	Prerequisites — System & Data Setup	S4 + BTP	■ Before all tests
TC-02	ATO SaaS triggers PO creation event	BTP — ATO SaaS	■ Testable NOW
TC-03	iFlow receives and transforms event → S4 OData API Int.	BTP Integration Suite	■ Testable NOW
TC-04	S4 creates PO — standard + ATO extension fields written	S4 — QJ6-001	■ Testable NOW
TC-05	Change Document records PO creation (CDHDR/CDPOS)	S4 — QJ6-001	■ Verified CE2608
TC-06	ATO field change in ME22N — Change Document updated	S4 — QJ6-001	■ Testable NOW
TC-07	Audit log — where is it stored and how to read it	S4 — QJ6-001	■ Testable NOW
TC-08	ATO field change log visible in ME22N/ME23N screen	S4 — QJ6-001	■ Open Issue — CE2608
TC-09	S4 publishes outbound event back to ATO SaaS	S4 → Event Mesh → BTP	■ Testable NOW
TC-10	PO cancellation from ATO — S4 marks item for deletion	BTP ATO → S4	■ Testable NOW

System & Package Reference

Item	Value	Notes
POC System	QJ6-001	S4 development system for CGTO/ATO POC
Package	CGTO_S4HC_POC	All ATO PO extension objects delivered here
CE2608 Deliverable	OData V4 PurchaseOrder API internal extensibility enabled	Done — no further action needed
CE2608 Deliverable	Change Document verified for PO fields	Done — CDHDR/CDPOS working

2702 Target	eSignature + Reauthentication integration in ME21N	Planned
ATO CDS View in CDPOS	ATO CDS view name recorded as TABNAME in CDPOS	Standard OBJECTCLAS = EINKBELEG used
Implicit Enhancement	Added in standard programme for ATO field description display	POC approach — acceptability to be confirmed with PO standard team

Prerequisites — System & Data Setup

Complete before running any test case

All test cases depend on these prerequisites being in place. Verify each item before starting TC-02.

#	Prerequisite	How to Verify	Owner
P1	Access to QJ6-001 S4 system granted	Login to QJ6-001 — confirm ME21N is accessible	Basis / Admin
P2	Mandatory GxP training completed	Training record logged — required before accessing QJ6-001	Developer / Tester
P3	ATO SaaS (BTP) access available	Login to ATO cloud portal — confirm specimen shipment screen is accessible	ATO team
P4	iFlow deployed in Integration Suite	BTP Integration Suite → Monitor → check PO create/update iFlow is active	Integration team
P5	Event Mesh subscription active	BTP Event Mesh → check ATO outbound PO event topic subscription is live	Integration team
P6	Test vendor exists in QJ6-001	ME22N → open any PO — confirm vendor master data present	MM Config
P7	Test material exists in QJ6-001	MM03 → confirm material for therapy type exists	MM Config
P8	SE16N access available for audit log verification	SE16N → open CDHDR — confirm access	Basis
P9	ATO extension tables exist in QJ6-001	SE11 → /ATO/S4_PO_HEADER_EXT — confirm table is Active	ATO Dev team
P10	Change Document object EINKBELEG includes ATO table	SE11 → Change Document Object → EINKBELEG → check ATO table is registered	ATO Dev team

02

ATO SaaS — Trigger PO Creation Event

System: BTP — ATO SaaS Portal | Testable NOW

Step	Action	Detail / Input
1	Open ATO Portal	Log in to ATO Cloud application (SaaS). Navigate to 'Manage Schedules' or 'Specimen Shipment' screen.
2	Prepare test data	Create or locate a CR ID (Collection Request) that requires a purchase order. Ensure Clinical Study ID, Protocol ID and Therapy Type are set: • Therapy Type: CELL_GENE • Clinical Study: CS-TEST-001 • Protocol: PROT-TEST-01 • Patient Ref: PAT-TEST-001 (autologous)
3	Set planning dates	In 'Manager Schedules' set Pickup Date and Due Date.
4	Trigger ERP document creation	Click the 'Trigger ERP Document Creation' button. Select the specimen shipment. Confirm the trigger.
5	Check outbound event	Navigate to BTP Event Mesh → Messages Inbox → find the outbound event. Event type: PurchaseOrder.Create Confirm payload contains: • Vendor, Material, Quantity, Delivery Date (standard fields) • TherapyType, ClinicalStudy, ProtocolRef, PatientRef (ATO fields)

Verify Where	What to Check
ATO Portal	Outbound event visible in Event Mesh with correct payload including ATO fields
BTP Event Mesh	Event published to topic — status = Published. Payload contains all ATO fields.
BTP Monitor	No error in event publishing — event consumed by iFlow subscriber

✓ Expected Result	ATO SaaS publishes a PO creation event to Event Mesh with standard + ATO extension fields in the payload.
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03

iFlow — Transform ATO Event → S4 OData API Call

System: BTP Integration Suite | Testable NOW

Step	Action	Detail / Input
1	Monitor iFlow execution	BTP Integration Suite → Monitor → All Integration Flows → filter by PO Create flow.
2	Check message processing	Find the message triggered by the TC-02 event. Status should be COMPLETED.
3	Verify schema mapping	Open the message log — confirm iFlow mapped: • ATO event schema → S4 PurchaseOrder OData V4 API schema • TherapyType, ClinicalStudy, ProtocolRef, PatientRef fields mapped correctly
4	Confirm S4 API call	In iFlow log → outbound call → confirm POST to: /sap/opu/odata4/sap/api_purchaseorder_2/srvd_a2x/sap/purchaseorder/0001/PurchaseOrder HTTP status = 201 Created

Verify Where	What to Check
Integration Suite Monitor	Message status = COMPLETED — no errors in processing log
iFlow Payload Log	ATO fields correctly mapped to OData request body
Outbound HTTP Log	S4 OData API responded with HTTP 201 and returned PO number

✓ Expected Result	iFlow successfully transforms ATO event schema to S4 OData V4 format and calls the PurchaseOrder API. Response confirms PO number created.
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04

S4 — PO Created with Standard + ATO Extension Fields

System: S4 — QJ6-001 | Testable NOW

Step	Action	Detail / Input
1	Navigate to PO	ME23N → enter PO number returned from TC-03 iFlow log. Or ME21N search by vendor and date to find the newly created PO.
2	Verify standard PO fields	Header: Vendor, Document Date, Purchase Org, Company Code Item: Material, Quantity, Delivery Date, Plant
3	Verify ATO extension tab	Look for 'Advanced Therapy Information' tab on PO Header. Confirm values: • Therapy Type = CELL_GENE • Clinical Study = CS-TEST-001 • Protocol Ref = PROT-TEST-01
4	Verify ATO item fields	PO Item → Advanced Therapy Information tab Confirm: Patient Ref = PAT-TEST-001
5	Verify extension tables directly	SE16N → /ATO/S4_PO_HEADER_EXT → filter EBELN = Confirm row exists with correct ATO field values. SE16N → /ATO/S4_PO_ITEM_EXT → filter EBELN + EBELP = Confirm row exists.

Verify Where	What to Check
ME23N	PO exists with correct standard fields. Advanced Therapy Information tab visible with ATO values.
SE16N → /ATO/S4_PO_HEADER_EXT	Row exists: EBELN=, THERAPY_TYPE=CELL_GENE, CLINICAL_STUDY=CS-TEST-001, PROTOCOL_REF=PROT-TEST-01
SE16N → /ATO/S4_PO_ITEM_EXT	Row exists: EBELN+EBELP=, PATIENT_REF=PAT-TEST-001

✓ Expected Result	PO created in S4 with all standard fields AND ATO extension fields persisted in /ATO/S4_PO_HEADER_EXT and /ATO/S4_PO_ITEM_EXT tables.
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Change Document Created on PO Creation (CDHDR + CDPOS)

System: S4 — QJ6-001 — Verified CE2608 | ■ Verified CE2608

Step	Action	Detail / Input
1	Check via ME23N	ME23N → open the PO from TC-04 → Menu: Environment → Changes A change log screen opens showing the creation record.
2	Check CDHDR directly	SE16N → CDHDR → filter: OBJECTCLAS = EINKBELEG OBJECTID = Note the CHANGENR value for use in subsequent steps.
3	Check CDPOS for standard fields	SE16N → CDPOS → filter CHANGENR = Expect rows for standard PO fields created: TABNAME = EKKO, FNAME = various standard fields
4	Check CDPOS for ATO fields	In same CDPOS filter result, look for rows where: TABNAME = /ATO/S4_PO_HEADER_EXT or ATO CDS view name FNAME = THERAPY_TYPE / CLINICAL_STUDY / PROTOCOL_REF VALUE_NEW = values set in TC-02

Verify Where	What to Check
SE16N → CDHDR	One row: OBJECTCLAS=EINKBELEG, OBJECTID=, USERNAME=, UDATE=today
SE16N → CDPOS standard	Rows with TABNAME=EKKO/EKPO, standard field names, VALUE_NEW populated
SE16N → CDPOS ATO	Rows with TABNAME=ATO table/CDS, FNAME=THERAPY_TYPE etc., VALUE_NEW=CELL_GENE etc.
ME23N → Environment → Changes	Change log screen shows PO creation entry with date and user

✓ Expected Result	CDHDR has one header row. CDPOS has rows for both standard PO fields AND ATO extension fields. This confirms Change Document is working for ATO fields — verified CE2608.
■ Open Issue	ATO field changes appear in CDPOS but the DESCRIPTION column is blank in ME22N/ME23N change log screen because ATO table name is not in standard PO change document object's description mapping. ATO team added implicit enhancement as workaround — acceptability to be confirmed with PO standard team.

06

User Changes ATO Field in ME22N — Change Document Updated

System: S4 — QJ6-001 | Testable NOW

Step	Action	Detail / Input
1	Open PO in change mode	ME22N → enter PO number from TC-04 → click Edit (pencil icon)
2	Change an ATO field	Navigate to 'Advanced Therapy Information' tab on PO Header. Change Therapy Type from CELL_GENE to PLASMA. Note: this is a REGULATED field — in 2702 this will trigger reauthentication.
3	Save the PO	Click Save. PO saved successfully (no reauthentication dialog in current state — 2702 pending).
4	Note the change timestamp	Record the exact time of save for cross-reference with CDHDR.

Verify Where	What to Check
SE16N → CDHDR	New row added: OBJECTCLAS=EINKBELEG, USERNAME=, UDATE=today, UTIME=save time. Note new CHANGENR.
SE16N → CDPOS	Filter new CHANGENR: row with TABNAME=ATO table, FNAME=THERAPY_TYPE, VALUE_OLD=CELL_GENE, VALUE_NEW=PLASMA
SE16N → /ATO/S4_PO_HEADER_EXT	THERAPY_TYPE field now = PLASMA (current live value updated)
ME23N → Environment → Changes	New change entry appears in change log for this PO

✓ Expected Result

CDHDR has a new row for this change. CDPOS has a row with VALUE_OLD=CELL_GENE and VALUE_NEW=PLASMA. This is the GxP audit trail — who changed what, from what value, to what value, when.

TC-07 — Where the Audit Log Is Stored and How to Read It

Complete audit trail reference for GxP compliance

7.1 Audit Log Storage Locations

Store	Table / Location	What It Contains	System
S4 Change Doc Header	CDHDR	PO number (OBJECTID), Change Doc number (CHANGENR), User (USERNAME), Date (UDATE), Time (UTIME), Object class (EINKBELEG)	QJ6-001
S4 Change Doc Item	CDPOS	CHANGENR (links to CDHDR), Table name (TABNAME), Field name (FNAME), Old value (VALUE_OLD), New value (VALUE_NEW)	QJ6-001
S4 ATO Current Values	/ATO/S4_PO_HEADER_EXT /ATO/S4_PO_ITEM_EXT	Latest current values of ATO fields only — NOT history. Query for current state.	QJ6-001

BTP eSignature (2702)	BTP eSignature SaaS	Signature ID, User, Timestamp, Reason Code, Credential hash. Linked to CHANGENR in CDHDR.	BTP — planned 2702
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7.2 How to Read the Complete Audit Trail

STEP 1 – Find all changes to a specific PO:

SE16N → CDHDR

Filter: OBJECTCLAS = EINKBELEG

OBJECTID = <PO number e.g. 4500000123>

Result: list of CHANGENR rows – one row per save event

each row shows: USERNAME, UDATE, UTIME

STEP 2 – For each CHANGENR, see what fields changed:

SE16N → CDPOS

Filter: CHANGENR = <value from CDHDR>

Result: one row per changed field

TABNAME = table where field lives (EKKO, /ATO/S4_PO_HEADER_EXT etc.)

FNAME = field name (THERAPY_TYPE, LIFNR etc.)

VALUE_OLD = value before change

VALUE_NEW = value after change

STEP 3 – See current live values of ATO fields:

SE16N → /ATO/S4_PO_HEADER_EXT

Filter: EBELN = <PO number>

Result: current values of THERAPY_TYPE, CLINICAL_STUDY, PROTOCOL_REF

STEP 4 – Display via ME23N (standard UI):

ME23N → open PO → Environment menu → Changes

Shows: standard PO field changes in formatted view

NOTE: ATO extension field changes may NOT show description text (open issue)

STEP 5 – After 2702: correlate with eSignature record:

Take CHANGENR from CDHDR

Find matching Signature ID in BTP eSignature SaaS admin console

Confirms: same user, same timestamp, reason code provided

Combined CDHDR + CDPOS + BTP Signature = complete 21 CFR Part 11 audit trail

7.3 CDPOS Key Fields — Quick Reference

Field	Meaning	Example Value
CHANGENR	Change Document Number — links CDHDR to CDPOS	0000012345
TABNAME	Table where the changed field lives	EKKO or /ATO/S4_PO_HEADER_EXT
FNAME	Field name that changed	THERAPY_TYPE
VALUE_OLD	Value BEFORE the change — the audit trail past	CELL_GENE
VALUE_NEW	Value AFTER the change — what it is now	PLASMA
CHNGIND	Change indicator: U=Update, I=Insert, D=Delete	U
TABKEY	Key of the changed record (PO number + item etc.)	4500000123001

ATO Field Change Log Visible in ME22N/ME23N Screen

System: S4 — QJ6-001 — OPEN ISSUE | ■ Open Issue

Step	Action	Detail / Input
1	Open PO change log	ME22N or ME23N → open PO from TC-04 → Environment → Changes
2	Look for ATO field changes	In the change log screen, look for rows where Therapy Type, Clinical Study, Protocol Ref appear as changed fields.
3	Check description column	ATO change rows may show with blank description in the 'Field' column because the ATO table name is not in the standard description mapping.
4	Check if implicit enhancement active	If ATO team's implicit enhancement is deployed, the description should show the ATO field name. Verify this is working.

Verify Where	What to Check
ME23N → Environment → Changes	Standard PO field changes visible with field labels. ATO field changes: check if THERAPY_TYPE, CLINICAL_STUDY, PROTOCOL_REF appear.
CDPOS cross-check	SE16N → CDPOS → CHANGENR → confirm ATO rows exist even if not displayed in ME23N

✓ Expected Result	CURRENT STATE: ATO field changes recorded in CDPOS but description blank in ME23N. Data IS captured — display is the gap. TARGET STATE: ATO fields appear with proper labels in ME22N/ME23N change log.
■ Open Issue	Resolution being investigated: BAdI 'Change Document Processing' (identified by Jiss, PO team) OR confirm if ATO implicit enhancement is sufficient. Action: joint call — ATO team (Allen Yuan) + PO team (Jiss) + Change Document team.

09

S4 Publishes Outbound Event Back to ATO SaaS

System: S4 → Event Mesh → BTP ATO SaaS | Testable NOW

Step	Action	Detail / Input
1	Make a change in ME22N	Change a standard or ATO field on the PO (e.g. delivery date or therapy type). Save.
2	Check S4 outbound event	S4 Event Mesh configuration → confirm PO change event was published. Event type: sap.s4.beh.purchaseorder.v1.PurchaseOrder.Changed.v1
3	Check Event Mesh	BTP Event Mesh → check the topic for PO change events — new message visible.
4	Check iFlow processing	Integration Suite → Monitor → PO Update iFlow → confirm COMPLETED status.
5	Verify ATO receives update	ATO Cloud portal → open the specimen shipment linked to this PO → confirm the changed field value is reflected on the ATO side.

Verify Where	What to Check
S4 Event Framework	Outbound event published after PO save
BTP Event Mesh	Message visible in PO change topic
Integration Suite	iFlow processed message with status COMPLETED
ATO SaaS Portal	Changed field visible on ATO specimen shipment — linkage confirmed

✓ Expected Result	PO change event flows from S4 → Event Mesh → iFlow → ATO SaaS. ATO side reflects the change — bidirectional sync confirmed.
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PO Cancellation from ATO — S4 Marks Item for Deletion

System: BTP ATO → S4 | Testable NOW

Step	Action	Detail / Input
1	Trigger cancellation in ATO	ATO Cloud portal → open specimen shipment → change status to CANCELLED.
2	ATO sends cancellation event	ATO publishes outbound event to Event Mesh. Event payload: PO number + cancellation instruction.
3	iFlow processes cancellation	Integration Suite iFlow transforms event → calls S4 PurchaseOrder API PATCH/DELETE.
4	Verify in ME23N	ME23N → open PO → item should show 'Mark for Deletion' indicator (L column = X).
5	Verify change document	SE16N → CDPOS → find new CHANGENR for this PO → row with FNAME = LOEKZ (deletion indicator), VALUE_OLD = space, VALUE_NEW = X

Verify Where	What to Check
ME23N	PO item shows deletion indicator (marked for deletion — NOT hard deleted)
ATO Portal	Specimen shipment status = CANCELLED — PO link shows deletion status
SE16N → CDPOS	New CDPOS row: FNAME=LOEKZ, VALUE_OLD=space, VALUE_NEW=X — audit trail preserved

✓ Expected Result

PO item marked for deletion in S4 when cancelled from ATO. Cancellation recorded in Change Document. ATO portal reflects updated status. NOTE: S4 does NOT hard-delete PO lines — only marks for deletion (GxP: data must be retained).

Audit Log — Complete Storage Reference

Where every piece of the GxP audit trail lives

COMPLETE AUDIT TRAIL PICTURE

1. ATO SaaS (BTP)

- Who triggered: user logged into ATO portal
- What: specimen shipment schedule confirmed → PO creation triggered
- When: ATO event timestamp
- Stored in: ATO Cloud application database (BTP)

2. Event Mesh (BTP)

- Event published: PurchaseOrder.Create / PurchaseOrder.Changed

- Payload: all field values including ATO extension fields

- Stored in: BTP Event Mesh message log

3. Integration Suite iFlow (BTP)

- Message processing log: input payload, output payload, HTTP status

- Schema mapping trace: ATO schema → S4 OData schema

- Stored in: BTP Integration Suite monitoring

4. S4 PO Tables (QJ6-001)

- EKK0: Standard PO header – current values

- EKP0: Standard PO items – current values

- /ATO/S4_PO_HEADER_EXT: ATO header fields – CURRENT values only

- /ATO/S4_PO_ITEM_EXT: ATO item fields – CURRENT values only

5. Change Document (QJ6-001) ← THE PRIMARY GxP AUDIT TRAIL

- CDHDR: WHO changed, WHEN changed, WHICH PO

- OBJECTCLAS=EINKBELEG, OBJECTID=PO#, USERNAME, UDATE, UTIME

- CDPOS: WHAT changed – one row per field

- TABNAME=EKK0 or /ATO/S4_PO_HEADER_EXT

- FNAME=field name (THERAPY_TYPE etc.)

- VALUE_OLD=before, VALUE_NEW=after

- ← Historical record – never overwritten

6. BTP eSignature SaaS (Planned 2702)

- Signature ID linked to CHANGENR in CDHDR

- User identity re-verified at moment of change

- Reason Code captured

- Combined with CDHDR+CDPOS = 21 CFR Part 11 compliant

KEY DISTINCTION:

Tables (EKK0, /ATO/S4_PO_HEADER_EXT): CURRENT state only

CDHDR + CDPOS:

HISTORY of all changes – never deleted

BTP eSignature:

PROOF of intentional authorisation

■ OPEN ISSUE: ATO field changes (THERAPY_TYPE, CLINICAL_STUDY, PROTOCOL_REF) ARE correctly recorded in CDPOS with correct TABNAME, FNAME, VALUE_OLD and VALUE_NEW. However the ME22N/ME23N change log DISPLAY screen does not show ATO field labels — the description appears blank. The data is captured but the display is the gap. This must be resolved before GxP compliance can be claimed for ATO field changes.